

Perceptron: Theory, Learning, and Boolean Applications

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Perceptron — Basic Model

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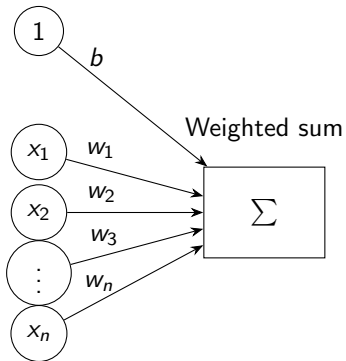
x_1

x_2

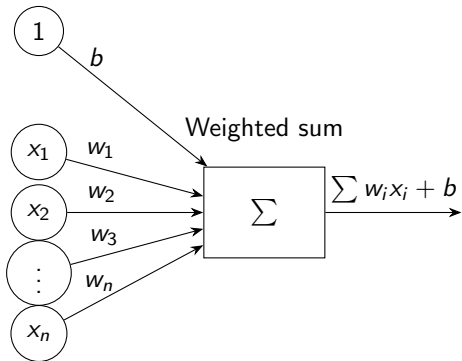
$\vdots$

x_n

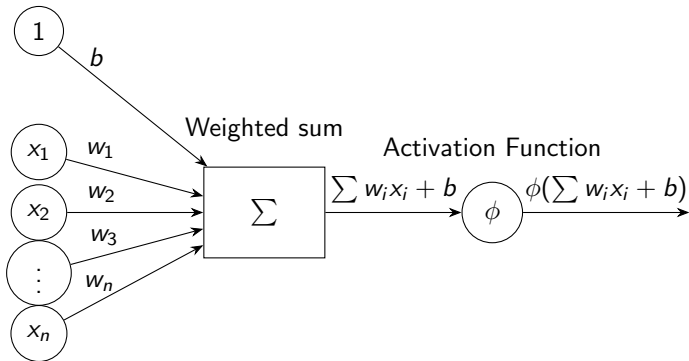
Perceptron — Basic Model



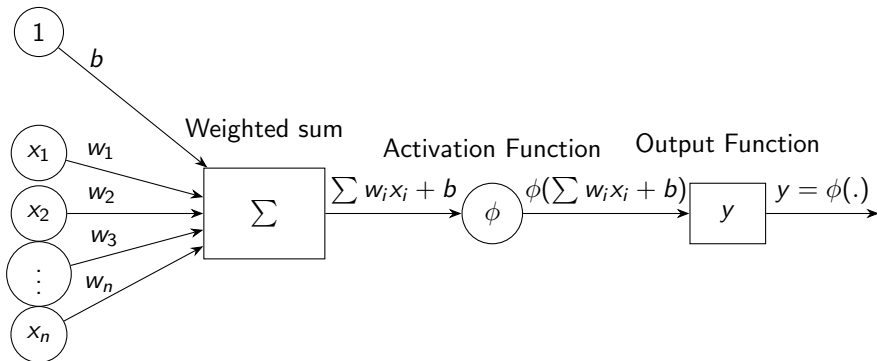
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Core Equation

Core equations

$$z = \mathbf{w} \cdot \mathbf{x} + b$$

$$y = \phi(z) = \begin{cases} 1 & z > 0 \\ 0 & z \leq 0 \end{cases}$$

$$\Delta w_i = \eta(t - y)x_i, \quad \Delta b = \eta(t - y)$$

Components

z = **Weighted Sum**

$\phi(\cdot)$ = **Activation Function**

Δw_i = **Weight differences**

Δb = **Bias differences**

η = **Learning rate**

Error Calculation Objective

Error and loss functions

- Classification error: $e = t - y$
- Perceptron criterion: $L_P(\mathbf{w}) = - \sum_{\text{mis}} t(\mathbf{w} \cdot \mathbf{x} + b)$
- Alternative (for analysis): $E = \frac{1}{2} \sum (t - y)^2$

Convergence Theorem

If data are linearly separable, perceptron converges in finite updates.
Bound on number of mistakes:

$$M \leq \left(\frac{R}{\gamma} \right)^2,$$

where $R = \max$ norm of inputs, $\gamma = \text{margin}$.

Applications and Limitations

Applications

- Binary classification (spam, signal thresholding)
- Basis for MLPs and feature detectors
- Used in linear classifiers and simple pattern recognition

Limitations

- Only linearly separable problems solvable
- Cannot model XOR/parity
- Discrete output, no probability interpretation
- Sensitive to feature scaling and noise

Boolean Functionality — Mathematical Depth

Linear separability of Boolean functions

A Boolean function $f : \{0, 1\}^n \rightarrow \{0, 1\}$ is linearly separable if $\exists \mathbf{w}, b : f(\mathbf{x}) = 1_{\{\mathbf{w} \cdot \mathbf{x} + b > 0\}}$.

Examples (2-input)

- AND: $w = (1, 1), b = -1.5$ separable
- OR: $w = (1, 1), b = -0.5$ separable
- XOR: not separable (no single hyperplane works)

Counting

Total Boolean functions: 2^{2^n}

Linearly separable counts: $n = 1 : 4, n = 2 : 14, n = 3 : 104$ (of 256 total)

Feature Maps for Nonlinear Problems

Mapping to higher dimension

Example (to make XOR linear):

$$\phi(x_1, x_2) = (x_1, x_2, x_1x_2)$$

Then $x_1 \oplus x_2 = x_1 + x_2 - 2x_1x_2$ (linear in ϕ).

Perceptron Learning Algorithm

Algorithm (Pseudo-code)

- ① Initialize weights w_i and bias b to small random values.
- ② For each training sample $(\mathbf{x}, t)$:
 - Compute $y = \phi(\mathbf{w} \cdot \mathbf{x} + b)$
 - Update: $w_i \leftarrow w_i + \eta(t - y)x_i, \quad b \leftarrow b + \eta(t - y)$
- ③ Repeat for all samples until convergence or max epochs.

Remarks

Learning rate η controls the step size. If data are linearly separable, weights converge in finite iterations (Perceptron Convergence Theorem).

Worked Example: Learning AND Gate

Training Data

x_1	x_2	t
0	0	0
0	1	0
1	0	0
1	1	1

Initialization

$w_1 = 0, w_2 = 0, b = 0,$
 $\eta = 1$

Iterations

- Sample (0,0): $y = 0, t = 0 \rightarrow$ no change
- Sample (0,1): $y = 0, t = 0 \rightarrow$ no change
- Sample (1,0): $y = 0, t = 0 \rightarrow$ no change
- Sample (1,1): $y = 0, t = 1 \rightarrow$
 $\Delta w_1 = 1, \Delta w_2 = 1, \Delta b = 1$

Final Model

$w_1 = 1, w_2 = 1, b = -1.5$ (adjust bias
for correct threshold)

Decision: $y = 1$ if $x_1 + x_2 > 1.5$, else 0.

Threshold and Bias in Perceptron

Bias (b) Definition

The bias allows the perceptron to **shift the decision boundary** away from the origin.

$$y = \phi(\mathbf{w} \cdot \mathbf{x} + b)$$

- If $b = 0$, the hyperplane passes through the origin.
- Positive b shifts the hyperplane **left/up**.
- Negative b shifts the hyperplane **right/down**.

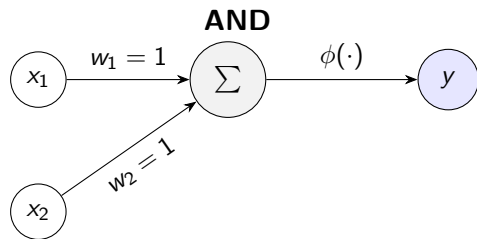
Threshold (θ) Concept

Alternatively, the perceptron can be written with a **threshold**:

$$y = \begin{cases} 1 & \text{if } \mathbf{w} \cdot \mathbf{x} \geq \theta \\ 0 & \text{otherwise} \end{cases}$$

Relationship: $b = -\theta$

Graphical Neuron: AND vs OR (Same $\theta = 2$)

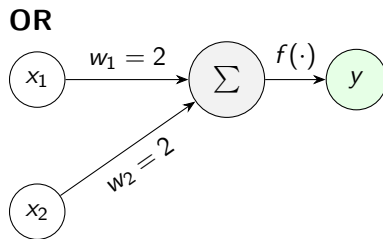


$$\theta = 2$$

$$\text{So, } b = -2$$

$$\begin{aligned}\sum_w &= w_1x_1 + w_2x_2 + b \\ &= x_1 + x_2 + (-2)\end{aligned}$$

$$\text{Output: } y = \phi(x_1 + x_2 - 2)$$



$$\theta = 2$$

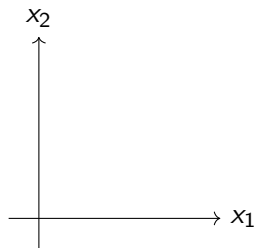
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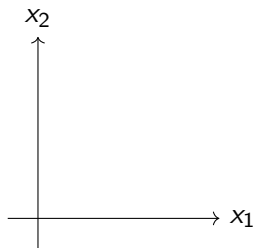
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AND and OR Gate Feature Maps with Hyperplane Separation

AND Gate

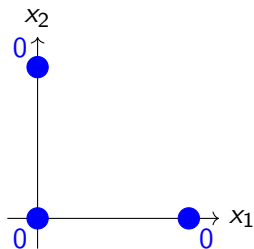


OR Gate

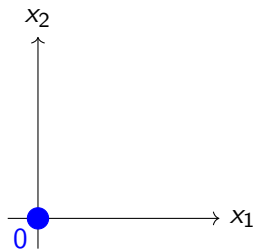


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AND Gate

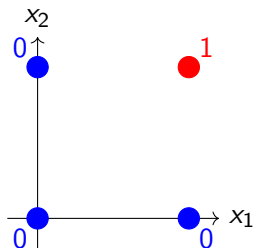


OR Gate

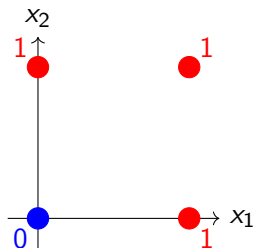


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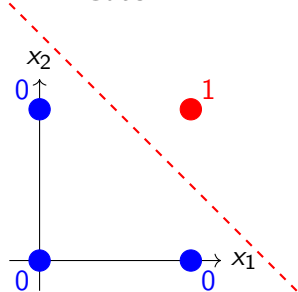


OR Gate



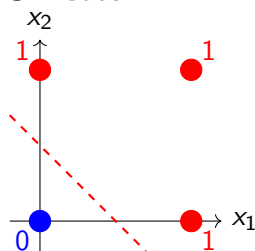
AND and OR Gate Feature Maps with Hyperplane Separation

AND Gate



Decision Boundary

OR Gate



Decision Boundary

Derivation of Perceptron Matrix Formula

1. Single perceptron (scalar form): For input vector $\mathbf{x} \in \mathbb{R}^n$, the output is:

$$y = f\left(\sum_{i=1}^n w_i x_i + b\right)$$

where w_i are weights, b is bias, and $f(\cdot)$ is the step function.

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2. **Vector form:** Using linear algebra, the same formula becomes:

$$y = f(\mathbf{w}^T \mathbf{x} + b), \quad \mathbf{w} = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

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3. **Multiple input samples:** Let there be m input samples $\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(m)}$:

$$\mathbf{X} = \begin{bmatrix} \left. \begin{array}{c} | \\ \mathbf{x}^{(1)} \\ | \end{array} \right| & \left. \begin{array}{c} | \\ \mathbf{x}^{(2)} \\ | \end{array} \right| & \dots & \left. \begin{array}{c} | \\ \mathbf{x}^{(m)} \\ | \end{array} \right| \end{bmatrix} \in \mathbb{R}^{n \times m}$$

Each column is a sample.

Derivation of Perceptron Matrix Formula

4. Linear combination for all samples:

$$\mathbf{Z} = \mathbf{W}^T \mathbf{X} \in \mathbb{R}^{1 \times m}$$

- $\mathbf{W}^T$ is $1 \times n$ - $\mathbf{X}$ is $n \times m$ - $\mathbf{Z}$ contains the weighted sum for each sample

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5. Adding bias: Bias must be added for each sample:

$$\mathbf{B} = [b \quad b \quad \dots \quad b] \in \mathbb{R}^{1 \times m}$$

$$\mathbf{Z} + \mathbf{B} = \mathbf{W}^T \mathbf{X} + \mathbf{B}$$

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6. Applying activation function: Finally, the perceptron output vector is:

$$\mathbf{y} = f(\mathbf{W}^T \mathbf{X} + \mathbf{B}) \in \mathbb{R}^{1 \times m}$$

Each element corresponds to the output for a sample.